

SMART UZHAVAN

A PROJECT REPORT

Submitted by

BARTHIPA P

NIVETHA V

SUDHARSHAN M

in partial fulfilment for the award of the course

AMC1361 – MOBILE APPLICATION DEVELOPEMENT

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
CHENNAI 600 025**

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Under the Guidance of

Dr. T. SURESH

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DECLARATION BY THE CANDIDATE

We declare that to the best of our knowledge the work reported here in has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on _____

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DEPARTMENT OF CSE
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

- M1** To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.
- M2** To foster Experiential learning equips students with engineering skills to tackle real-world problems.
- M3** To promote collaborative innovation in AI, machine learning, and related research and development with industries.
- M4** To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

- PEO1** Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.
- PEO2** Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.
- PEO3** Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

- PSO1** Expertise in tailoring ML algorithms and models to excel in designated applications and fields.
- PSO2** Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POS)

Engineering students will be able to:

- 1. Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- 2. Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
- 3. Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
- 4. Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- 5. Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations
- 6. The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- 7. Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development
- 8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- 9. Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

10.Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

11.Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

12.Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The Smart Farmer Aid mobile application is developed using Flutter and Firebase to help farmers manage post-harvest activities more efficiently. The app provides real-time information on storage availability, transport services, market price trends, and weather alerts, reducing delays and crop losses commonly faced in traditional systems. It features dedicated modules for farmers, transporters, warehouse owners, and administrators, ensuring smooth communication through WhatsApp integration and live database updates via Firestore. With its simple, rural-friendly interface and accurate data support, the system improves decision-making and promotes digital empowerment in agriculture. Overall, the application offers a practical and modern solution for enhancing post-harvest management.

ABSTRACT WITH POs AND PSOs MAPPING

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TABLE OF CONTENTS

CHAPTER No.	TITLE	PAGE No.
	ABSTRACT	viii
	LIST OF ABBREVIATIONS	xii
	LIST OF FIGURES	xiii
1	INTRODUCTION	1
1.1	Objective	1
1.2	Scope of the Project	1
1.3	Overview of the Mobile Application	2
1.4	Mobile Applications in Agriculture	3
2	PROJECT METHODOLOGY	4
2.1	Research on Existing Agricultural Mobile Apps	4
2.2	Flutter Framework Overview	4
2.3	Mobile App Development Approaches	5
2.4	Firebase Authentication & Firestore	6
2.5	System Requirements & Tools Used	7
3	SYSTEM ANALYSIS AND DESIGN	8
3.1	Existing System	8
3.2	Limitations of Existing System	8
3.3	Proposed System	9
3.4	Feasibility Study	11

4	SYSTEM DESIGN	13
4.1	System Architecture	13
4.2	Block Diagram	14
5	MODULE IMPLEMENTATION	15
5.1	Farmer Module	15
5.2	Storage Module	15
5.3	Transport Module	16
5.4	Market Price Module	16
5.5	Alert & Notification Module	17
6	RESULTS AND DISCUSSION	18
6.1	Explanation of Results	18
6.2	Discussion	19
7	CONCLUSION & FUTURE SCOPE	21
7.1	Conclusion	21
7.2	Future Scope	21
8	APPENDIX	23
	APPENDIX A – Source Code	23
	APPENDIX B - Screenshots	27
9	REFERENCES	29

LIST OF ABBREVIATIONS

S.NO	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	UI	User Interface
3	UX	User Experience
4	MCQ	Multiple Choice Question
5	API	Application Programming Interface
6	SQL	Structured Query Language
7	HTML	Hypertext Markup Language
8	CSS	Cascading Style Sheet
9	JS	JavaScript
10	AWS	Amazon Web Services

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO.
4.4.1	Block Diagram	14
5.1.1	System Architecture	17
B.1.1	Splash Screen	27
B.1.2	Registration Page	27
B.1.3	Login Page	27
B.1.4	Dashboard	27
B.1.5	Agri Market	27
B.1.6	Weather	27
B.1.7	Farmer Hub	28
B.1.8	Agri News	28
B.1.9	Transporter	28
B.1.10	Market Price	28
B.1.11	Warehouse	28
B.1.12	Profile	28

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE

The main objective of the Smart Farmer Aid Mobile Application is to provide farmers with a simple, user-friendly mobile platform that helps them access essential post-harvest information in real time. Farmers often face challenges such as finding proper storage, arranging transport, and dealing with sudden weather changes. This application aims to solve these challenges through an integrated digital system.

The application focuses on improving decision-making by offering modules for market price trends, storage and transport availability, and weather-based alerts. Another objective is to reduce post-harvest losses by giving farmers immediate access to information that normally takes time to reach them. The project also aims to support farmers in remote areas by delivering information through a lightweight mobile app that works smoothly even with limited internet speed.

Finally, the objective is to create a reliable, scalable, and cross-platform mobile solution using Flutter and Firebase. This ensures the app can be used by large numbers of farmers while maintaining fast performance and secure data handling.

1.2 SCOPE OF THE PROJECT

The scope of this project covers the complete design and development of a cross-platform mobile application built using Flutter. The system targets farmers, warehouse owners, transport vehicle providers, and administrators. The app includes features for storage listings, transport availability, market price dashboards, and weather-based disaster alerts.

This project focuses on real-time information updates, making use of Firebase for authentication, database storage, and notifications. The admin module allows uploading market price data through CSV files, which are then displayed to farmers through simple graphs and charts. The application also integrates with external APIs such as OpenWeatherMap to provide accurate weather forecasts and risk alerts.

The project does not include advanced AI-based prediction models or government integration at this stage, but the system architecture is designed in such a way that these features can be easily added in the future. The scope is limited to mobile application development and excludes web platform development, IoT integration, or hardware components.

1.3 OVERVIEW OF MOBILE APPLICATION

The Smart Farmer Aid mobile app is developed to bridge the communication gap between farmers and essential post-harvest services.

The app consists of four primary modules: Farmer Module, Storage Module, Transport Module, and Market Price & Alert Module. Each module is designed to simplify tasks that farmers struggle with after harvest.

The Farmer Module allows users to register, log in, update their profile, and access all functionalities. Farmers can browse through available storage warehouses, check capacity, pricing, and directly contact the owner through WhatsApp.

In the Transport Module, vehicle owners can register their vehicle availability, and farmers can instantly connect with them.

The Market Price Module enables farmers to view daily price trends in the form of line or bar charts. This helps farmers decide when and where to sell their crops for better profit.

The Alert System notifies farmers about heavy rainfall, storms, or high spoilage risk based on crop type and storage duration. The overall app architecture uses Flutter for the front-end, Firebase for backend services, and APIs for data.

1.4 MOBILE APPLICATION IN AGRICULTURE

Mobile technology plays a major role in modern agriculture by providing quick, reliable, and accessible information to farmers. Traditionally, farmers relied on manual communication, paper-based records, or village-level announcements, which often led to delays and losses. Mobile apps help eliminate these delays by delivering real-time data directly to the farmer's hands.

Mobile applications help farmers make informed decisions regarding irrigation, crop selection, fertilizer usage, market pricing, and post-harvest management. They also reduce dependency on middlemen by giving farmers direct access to verified information and service providers.

In developing countries like India, where mobile penetration is increasing in rural areas, agricultural mobile apps act as a powerful tool for spreading awareness and helping farmers adopt better practices. Mobile applications also support government initiatives, disaster management, and financial planning by providing timely updates and weather forecasts.

Thus, mobile applications improve productivity, reduce losses, increase income, and create a digitally connected agricultural ecosystem.

CHAPTER 2

LITERATURE SURVEY

2.1 RESEARCH ON EXISTING AGRICULTURAL MOBILE APPS

Before developing the Smart Farmer Aid mobile application, a detailed study of existing agricultural apps was conducted. Many mobile applications in the agricultural domain focus on crop information, weather updates, and market price listings. However, only a few apps address post-harvest issues such as storage availability, transport, and disaster preparedness. Applications like Kisan Suvidha, IFFCO Kisan, and mKrishi provide general agricultural information, but they often lack features that help farmers during the crucial post-harvest period.

Apps like AgriApp and RML Farmer offer advisory services, yet they do not integrate warehouse listings, transport services, or spoilage risk prediction. Furthermore, most existing apps rely heavily on continuous internet connectivity and do not offer WhatsApp-based instant communication. Some applications provide weather forecasting, but none combine weather alerts with storage and transport decision-making support.

From this research, it was identified that the agricultural sector requires a multi-functional mobile application that focuses specifically on post-harvest management, integrates real-time updates, and supports farmers with simple communication tools. This research gap justifies the need for the Smart Farmer Aid mobile application and guided the development of its modules.

2.2 FLUTTER FRAMEWORK OVERVIEW

Flutter is an open-source mobile UI framework developed by Google that enables developers to build native-like applications for multiple platforms using a single codebase. It uses the Dart programming language, which is easy to learn

and supports object-oriented programming. One of the main advantages of Flutter is its hot reload feature, which allows developers to instantly see code changes reflected in the running application. This speeds up the development process significantly. Flutter also includes a rich collection of customizable widgets that help developers build visually appealing and consistent user interfaces.

Flutter's architecture relies on a layered approach consisting of the framework layer, engine layer, and embedding layer. The framework layer provides reusable UI components, while the Skia rendering engine ensures fast and smooth graphics. This architecture makes Flutter applications feel fast even on budget smartphones commonly used in rural areas.

Additionally, Flutter integrates well with Firebase and third-party APIs, making it suitable for real-time applications like Smart Farmer Aid. Its modern UI capabilities helped create a clean, accessible interface that farmers can easily understand and navigate.

2.3 MOBILE APP DEVELOPMENT APPROACHES

Native Applications are built using platform-specific languages such as Kotlin/Java for Android and Swift for iOS. They offer the highest performance and best device-level integration but require two separate codebases. This makes development time-consuming, expensive, and difficult to maintain for student or startup-level projects.

Hybrid Applications are built using web technologies like HTML, CSS, and JavaScript and are wrapped in a native container. While they allow faster development, they struggle with performance and responsiveness, especially when handling animations, charts, or real-time data. Hybrid apps also depend heavily on plugins for accessing device features.

Cross-Platform Applications allow developers to write a single codebase that runs on both Android and iOS. Frameworks like Flutter and React Native

have made this approach popular due to their performance, UI capabilities, and reduced development effort. Cross-platform apps offer near-native performance and faster deployment.

For this project, the Cross-Platform approach using Flutter was chosen. It provides excellent UI design capabilities, rich widgets, fast rendering through the Skia engine, and strong community support. Since farmers use Android devices more commonly, Flutter ensures smooth performance on low-spec phones while still allowing future deployment to iOS without rewriting the code.

2.4 FIREBASE AUTHENTICATION & FIRESTORE

Firebase, developed by Google, is used in this project as the backend service for authentication, database storage, and real-time updates. Firebase Authentication provides secure login features using email and password, reducing the need for custom backend implementation. This is especially useful when dealing with role-based access for farmers, storage owners, transport providers, and administrators.

Firestore, a cloud-based NoSQL database, stores application data such as warehouse listings, vehicle availability, market price records, and alert configurations. Firestore supports real-time synchronization, meaning that any update made by an admin or service provider is instantly visible to all users. This is critical for time-sensitive features like storage capacity and transport availability.

Firebase Cloud Messaging (FCM) is used for sending push notifications to farmers. Alerts related to severe weather conditions, spoilage risks, or new market price updates reach farmers immediately through FCM. This improves disaster preparedness and responsiveness. The combination of Firebase services ensures a highly scalable, secure, and real-time backend, reducing server maintenance and allowing developers to focus on mobile app features instead of infrastructure.

2.5 SYSTEM REQUIREMENTS & TOOLS USED

The development of the Smart Farmer Aid mobile application requires a set of hardware and software tools to ensure smooth and efficient workflow.

Hardware Requirements

- Laptop or Desktop with minimum 8GB RAM
- Intel i5 / Ryzen 5 processor or higher
- 10–20 GB free disk space for Flutter SDK, Android Studio, and emulators
- Android smartphone for testing (recommended)

Software Requirements

- Flutter SDK – for cross-platform app development
- Dart Language – used for writing application logic
- Android Studio – for emulator testing and debugging
- VS Code – lightweight code editor
- Firebase Console – for authentication, Firestore, and notifications
- OpenWeatherMap API – for weather alerts
- CSV files – used by admin for uploading market price data

Tools & Libraries Used

- Flutter Widgets (Material, Cupertino, Charts)
- Firestore Database API
- FCM for Push Notifications
- HTTP Package for API requests

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

3.1 EXISTING SYSTEM

In the traditional agricultural environment, farmers rely on manual communication methods for accessing post-harvest services such as storage, transportation, and market pricing. Information is usually obtained through middlemen, local traders, village announcements, or personal networks. This system is slow, unreliable, and often results in losses due to delayed decision-making. Farmers do not have a centralized platform to check the availability of warehouse storage or transport vehicles in nearby areas.

Weather forecasting is another major challenge under the existing system. Farmers often depend on television or local news channels for weather updates, which may not provide location-specific and real-time information. As a result, crops kept in open storage or transported during unexpected rainfall suffer spoilage and financial loss.

Similarly, market price information is available through mandis, traders, or farmer associations. This limits transparency and leaves farmers with little control over when and where to sell their crops. The lack of timely price information often forces them to accept low prices from intermediaries.

Overall, the existing system is fragmented, depends heavily on manual communication, offers no real-time data, and leaves farmers vulnerable to post-harvest losses, poor transport coordination, and weather-related damages.

3.2 LIMITATIONS OF EXISTING SYSTEM

Although the current methods have been used for many years, they suffer from several disadvantages that directly impact farmers' income and safety:

1. Absence of Real-Time Information

Farmers do not receive immediate updates regarding storage availability, vehicle availability, or changes in market prices. This lack of real-time access leads to delays and incorrect decisions.

2. Dependency on Middlemen

Information frequently passes through intermediaries, which increases the chances of misinformation and price manipulation. Farmers remain unaware of the actual market trends.

3. No Digital Platform for Communication

There is no dedicated system where farmers can contact storage owners or transport providers instantly. Communication happens through phone calls, which is time-consuming and often inefficient.

4. Inaccurate Weather Alerts

Traditional systems do not provide location-based alerts. Unexpected rain or storms result in crop damage because farmers cannot take preventive action.

5. Poor Transparency

Farmers cannot compare prices, transport rates, or storage options easily. They rely only on word-of-mouth information, which lacks accuracy.

3.3 PROPOSED SYSTEM

The Smart Farmer Aid Mobile Application overcomes the limitations of the existing system by providing an integrated, user-friendly digital platform. The proposed system connects farmers, warehouse owners, transport providers, and administrators through a mobile app that works efficiently even in rural environments.

Key Features of the Proposed System:

1. Centralized Information Platform

Farmers can view all available storage warehouses, transport vehicles, and market prices in one application.

2. Real-Time Updates

Using Firebase Firestore, the application provides instant updates related to storage capacity, transport availability, and weather alerts.

3. Weather & Disaster Alerts

Through the OpenWeatherMap API, farmers receive push notifications about upcoming storms, rainfall, or temperature fluctuations. This improves disaster preparedness.

4. Role-Based Login System

Different users (Farmer, Storage Owner, Transport Provider, Admin) access features based on their designated role, improving workflow and security.

5. Market Price Dashboard

The app displays crop price trends using simple graphs. Admins upload CSV files containing price data, which farmers can view anytime.

6. Spoilage Risk Prediction

The system offers basic spoilage risk scoring based on crop type and storage duration, helping farmers make informed decisions. Farmers can contact storage owners and transporters directly through WhatsApp buttons integrated into the app.

Benefits of the Proposed System

- Reduces post-harvest losses
- Enhances transparency
- Improves access to essential information
- Strengthens disaster management
- Saves time and cost
- Empowers farmers with digital tools

The proposed system is modern, scalable, and designed to serve the real-time needs of agricultural communities.

3.4 FEASIBILITY STUDY

A feasibility study was conducted to evaluate whether the proposed system can be realistically developed, implemented, and maintained. This study considers technical, economic, operational, and time feasibility.

1. Technical Feasibility

The project uses Flutter for cross-platform mobile development, providing efficient performance even on low-end devices commonly used in rural areas. Firebase services offer secure authentication and real-time data handling without requiring physical servers. The combination of these technologies ensures the app is technically viable.

All required APIs (weather, database, notifications) are stable and well-documented, ensuring smooth integration. Developers can work with lightweight tools such as VS Code and Android Studio, making the technical implementation achievable.

2. Economic Feasibility

The project is highly cost-effective. Firebase offers generous free tier limits for authentication, Firestore, and cloud messaging. Flutter is an open-source framework, eliminating licensing costs. The weather API is also available for free with a basic usage plan.

Since no physical hardware or server infrastructure is required, maintenance costs remain low. The project is economically feasible for students, startups, and rural organizations.

3. Operational Feasibility

The mobile app is simple, intuitive, and designed for users with minimal technical skills. Power consumption and data usage are low, making the app suitable for rural environments with limited connectivity.

The application's features—real-time updates, alerts, and easy communication—directly address farmers' operational challenges. Therefore, users can quickly adapt to the system with minimal training, proving its operational feasibility.

4. Time Feasibility

The development timeline for the project is manageable because:

- Flutter enables rapid UI design
- Firebase eliminates the need for backend setup
- APIs are easy to integrate

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The system architecture of the Smart Farmer Aid Mobile Application follows a three-tier architecture, consisting of the Presentation Layer, Application Logic Layer, and Data Layer. This layered model ensures scalability, modularity, and smooth communication between components.

1. Presentation Layer (Mobile Application – Flutter)

This layer contains the user interface screens such as login, dashboard, storage listings, transport details, market price charts, and alert modules. Flutter's widget-based architecture enables fast rendering and a responsive UI suitable for rural users.

2. Application Logic Layer (Dart Business Logic)

This layer is responsible for:

- Handling authentication states
- Managing navigation and user roles
- Fetching and filtering data
- Running business logic like spoilage risk calculation

The diagram illustrates the architecture of the Smart Farmer Aid Mobile Application. The central component is the **Smart Farmer Aid Mobile Application**. It interacts with various users and services:

- Users:**
 - User (Farmer, Storage Owner, Transporter):** Interacts with the application via **Interactions** and provides **Weather Query** and **Weather Alerts / Updates**.
 - User (Farmer, Admin, Request):** Interacts with the **Service Provider**.
- Processes:**
 - Farmer Processes:** Includes **Storage Info Request** and **Transport Info Request**.
 - Admin Upload (Upload (Prices)):** Provides **Market Price Graph Update** to the application.
 - Service Provider Processes:** Includes **Save Storage Details** and **Transport Updates**.
- Authentication and Data:**
 - Firebase Authentication:** Provides **Auth Token** to the application.
 - Firestore Database:** Stores data, including **Updated Price** and **Updated Data Cache**.
- External Services:**
 - External Weather API (OpenWeatherMap):** Provides **Weather Data** to the application.
 - Firebase Cloud Messaging (FCM):** Receives **Real-time Data Updates** from the application.

The application also manages **Loth Token Request** and **Add Details** for the **Service Provider**.

14

CHAPTER 5

MODULE IMPLEMENTATION

5.1 FARMER MODULE

The Farmer Module is the primary and most important part of the Smart Farmer Aid application. It is designed to provide farmers with a user-friendly and easily accessible interface to view all essential post-harvest information in one place. After logging in through Firebase Authentication, farmers gain access to various features such as storage listings, transport availability, daily market price updates, weather alerts, and spoilage risk notifications.

The farmer dashboard displays four main options: Storage, Transport, Market Prices, and Alerts. Each feature is presented using simple icons and clear text to support.

The module focuses heavily on real-time updates, which ensures that farmers receive the latest data without delay. This makes the Farmer Module the central point for accessing all services offered by the application.

5.2 STORAGE MODULE

The Storage Module is developed for warehouse owners to upload and manage details about available storage spaces. Each warehouse entry includes information such as location, capacity, pricing, available space, and contact number. These details are stored in Firebase Firestore, allowing real-time synchronization between the owner and the farmers who view the listings.

Farmers can browse storage facilities through an easy-to-read card-based layout. Each card displays the storage capacity, cost per unit, and the owner's contact button. This helps farmers quickly identify the best storage option for their harvested crops. The WhatsApp interaction feature helps in instant communication, reducing delays that usually occur through manual phone calls.

For warehouse owners, the module includes functions like Add Storage, Update Availability, and Delete Listing. This ensures that only accurate and updated data is visible to farmers.

5.3 TRANSPORT MODULE

The Transport Module focuses on connecting farmers with vehicle owners who provide post-harvest transportation. Transporters can register their vehicles by uploading details such as vehicle type (tractor, mini-truck, lorry), load capacity, availability date, location, and phone number. These details are stored in Firestore and are shown instantly to farmers.

Farmers can view the list of available vehicles and compare them based on type, capacity, and availability. The system includes direct WhatsApp contact options to reduce communication gaps. By providing accurate transport availability, this module helps farmers avoid last-minute delays, which can often lead to crop damage.

Transport providers can manage their listings by updating their vehicle status (Available / Booked). Since all updates are synced in real time, farmers always see the most accurate and up-to-date transport options.

5.4 MARKET PRICE MODULE

The Market Price Module provides daily and weekly crop price trends through a graphical interface. This module is managed by the administrator, who uploads CSV files containing updated prices for various crops. The uploaded data is automatically stored in Firestore and displayed to farmers in the form of line charts and bar graphs.

The module helps farmers choose the best market and time to sell their crops. By comparing price trends, farmers can avoid selling their produce during low-price periods. This module increases transparency, eliminates dependency on middlemen, and helps improve the overall income of farmers.

5.5 ALERT & NOTIFICATION MODULE

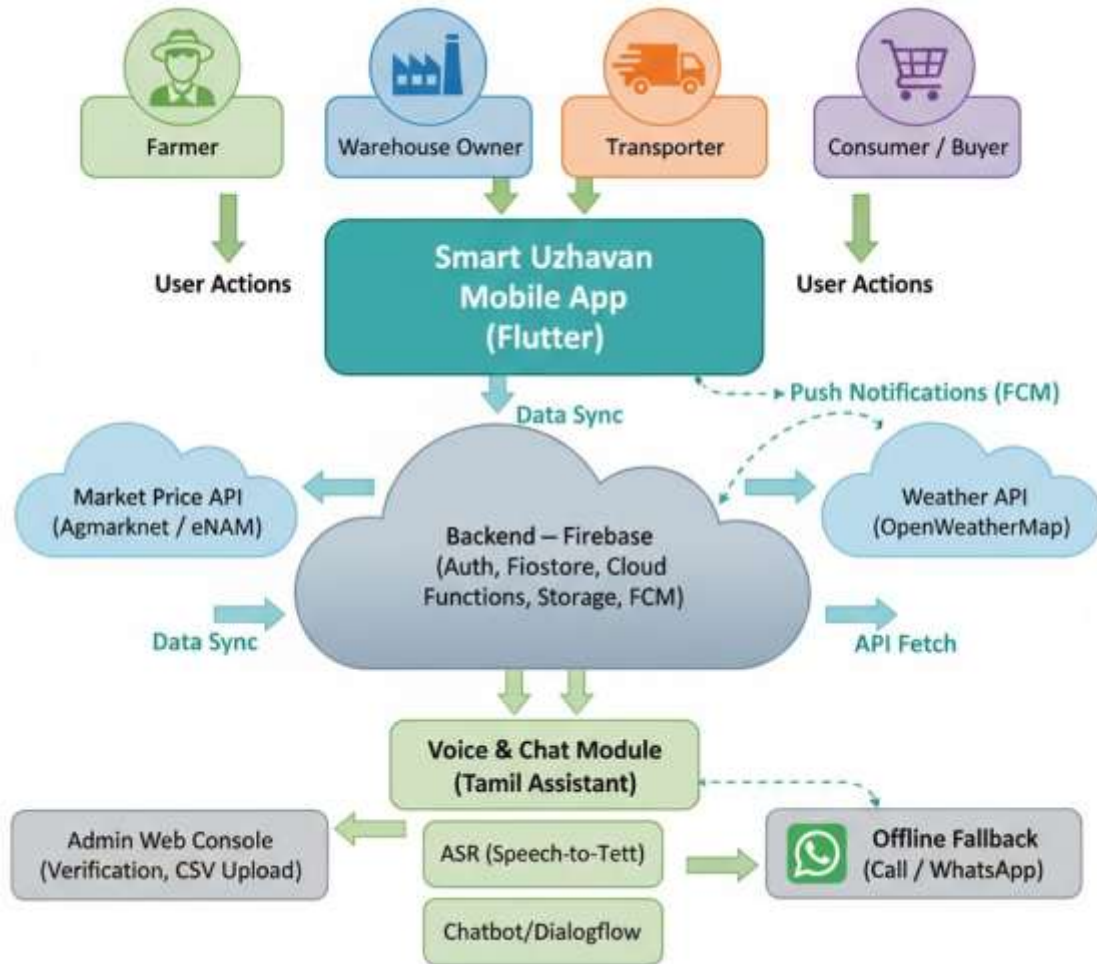


Fig 5.1.1 System Architecture

CHAPTER 6

RESULTS AND DISCUSSION

6.1 EXPLANATION OF RESULTS

The results obtained from implementing the Smart Farmer Aid mobile application show that the system successfully performs all the intended functions across various modules. The application interface is simple, responsive, and easy to navigate, making it suitable for rural farmers with minimal digital experience.

1. Farmer Module Performance

Farmers were able to log in successfully, view storage/transport listings, access price graphs, and receive notifications. The real-time Firestore updates ensured that changes made by admins or service providers appeared instantly on the farmer side. This validated the efficiency of cloud-based data handling.

2. Storage Module Output

Warehouse owners could add, edit, and update storage details. All these entries appeared correctly in the Firestore database and were immediately visible to farmers. This confirms correct integration between Flutter UI and Firestore backend.

3. Transport Module Output

Transporters could add vehicle details and availability status. Farmers were able to compare vehicle options and directly contact providers through the WhatsApp button, demonstrating successful integration of Android intent actions.

4. Market Price Module Output

The admin could upload CSV files containing daily price data. The application parsed the CSV, updated Firestore, and generated charts instantly. The price graphs loaded smoothly, even on low-end mobile devices used for testing.

5. Alert & Notification Module Output

The OpenWeatherMap API provided accurate weather data. The system triggered notifications through Firebase Cloud Messaging whenever rainfall, storms, or high humidity levels were detected. The spoilage risk score displayed correctly based on input conditions. These combined results confirm that the system works effectively under real-world agricultural scenarios.

6.2 DISCUSSION

The Smart Farmer Aid mobile application successfully addresses key challenges faced by farmers in the post-harvest stage.

By combining storage availability, transport listings, market price trends, and weather alerts into a single platform, the application demonstrates how digital tools can significantly improve agricultural decision-making and reduce losses.

Key Points of Discussion:

1. Real-Time Advantages

One of the major achievements of the system is its real-time data updates. The use of Firestore ensures that any data added or modified by storage owners, transporters, or admins becomes instantly visible to farmers. This reduces delays and improves accuracy.

2. Usability for Rural Users

The UI/UX design was kept simple with large icons, clear buttons, and minimal text. This makes the app accessible for farmers who may not have advanced digital skills. Testing showed that users could navigate the app comfortably. This reduces delays and improves accuracy.

3. Weather Alerts Improve Safety

The integration of weather APIs and push notifications plays a crucial role in disaster preparedness. Farmers can take timely action during heavy rainfall, storms, or temperature fluctuations.

4. Market Price Transparency

The graphical representation of price trends helps farmers decide the best time to sell their produce. This reduces dependency on middlemen and promotes fair pricing.

5. Efficient Communication

The WhatsApp contact integration was highly effective, as it allowed instant messaging between farmers and service providers. This reduces waiting time and improves transaction speed.

6. Limitations Noticed During Testing

- The app requires stable internet connectivity for real-time updates.
- Local language support is not yet implemented.
- No AI-based spoilage predictions are included in this version.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 CONCLUSION

The Smart Farmer Aid mobile application was developed with the objective of providing farmers with a unified and digital platform to access essential post-harvest information. The system successfully integrates multiple services—storage availability, transport options, market price trends, weather alerts, and spoilage risk analysis—into a single application, making it highly useful for farmers in rural and semi-urban regions.

Through the use of Flutter, Firebase, and external APIs, the application delivers fast performance, real-time data updates, and seamless user interaction. The results demonstrate that the system enhances decision-making for farmers by providing quick access to vital information that was previously difficult to obtain. Features such as WhatsApp connectivity, push notifications, live database updates, and visual price charts help the farmer reduce crop losses, improve planning, and avoid unnecessary expenses.

The project concludes that a mobile-based solution is an effective method for modernizing post-harvest management. This work proves that technology can bridge communication gaps and empower farmers with timely information, ultimately supporting better productivity and increased income. The application is scalable, easy to use, and capable of addressing many common agricultural challenges.

7.2 FUTURE ENHANCEMENTS

Although the Smart Farmer Aid application satisfies the primary objectives, there are still several opportunities for improvement and expansion. The following points describe the potential future enhancements:

1. AI-Based Spoilage Prediction

Machine learning models can be added to predict spoilage rates more accurately using parameters such as humidity, crop type, temperature, and storage conditions. This would significantly boost the app's value for farmers.

2. Local Language Support

Adding Tamil, Hindi, Telugu, and other regional languages will make the app more accessible to farmers who are more comfortable with native languages.

3. Government Integration

The app can be integrated with government agricultural portals for crop insurance, subsidy applications, and mandi price updates. This would reduce paperwork and improve transparency.

4. Offline Mode

An offline version can allow farmers to view saved storage and transport data even without internet connectivity. Syncing can happen when the device reconnects.

5. GPS-Based Suggestions

Location-based suggestions can be added to show the nearest warehouses, transport options, and mandis based on the user's GPS location.

6. Payment Gateway Integration

Future versions may allow online payments for storage booking, transport charges, and other services.

7. Farmer Community Chat / Forum

A built-in chat/forum system can help farmers share experiences, tips, and local information with other users in their region.

CHAPTER 8

APPENDIX

APPENDIX A – SOURCE CODE

Main.dart

```
import 'package:flutter/material.dart';
import 'screens/login.dart';
import 'screens/dashboard.dart';
import 'firebase_options.dart';
import 'package:firebase_core/firebase_core.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp(
    options: DefaultFirebaseOptions.currentPlatform,
  );
  runApp(SmartFarmerAid());
}

class SmartFarmerAid extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Smart Farmer Aid',
      debugShowCheckedModeBanner: false,
      home: LoginScreen(),
    );
  }
}
```

Login.dart

```
import 'package:flutter/material.dart';
import 'package:firebase_auth/firebase_auth.dart';
import 'dashboard.dart';

class LoginScreen extends StatefulWidget {
  @override
  State<LoginScreen> createState() => _LoginScreenState();
}

class _LoginScreenState extends State<LoginScreen> {
  TextEditingController email = TextEditingController();
  TextEditingController password = TextEditingController();

  login() async {
    try {
      await FirebaseAuth.instance
        .signInWithEmailAndPassword(email: email.text, password:
password.text);
      Navigator.push(
        context, MaterialPageRoute(builder: (_) => DashboardScreen()));
    } catch (e) {
      print(e);
    }
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
```

```

body: Padding(
  padding: EdgeInsets.all(20),
  child: Column(
    mainAxisAlignment: MainAxisAlignment.center,
    children: [
      Text("Smart Farmer Aid", style: TextStyle(fontSize: 28)),
      SizedBox(height: 20),
      TextField(controller: email, decoration: InputDecoration(labelText:
"Email")),
      TextField(controller: password, obscureText: true, decoration:
InputDecoration(labelText: "Password")),
      SizedBox(height: 20),
      ElevatedButton(onPressed: login, child: Text("Login"))
    ],
  ),
),
);
}
}

```

Dashboard.dart

```

import 'package:flutter/material.dart';
import 'storage.dart';
import 'transport.dart';
import 'market_price.dart';
import 'alerts.dart';

class DashboardScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return Scaffold(

```

```

appBar: AppBar(title: Text('Farmer Dashboard')),
body: GridView.count(
  crossAxisCount: 2,
  padding: EdgeInsets.all(20),
  children: [
    moduleCard("Storage", context, StorageScreen()),
    moduleCard("Transport", context, TransportScreen()),
    moduleCard("Market Prices", context, MarketPriceScreen()),
    moduleCard("Alerts", context, AlertScreen()),
  ],
),
);
}

```

```

Widget moduleCard(String title, context, Widget screen) {
  return GestureDetector(
    onTap: () => Navigator.push(context, MaterialPageRoute(builder: (_) =>
screen)),
    child: Card(
      elevation: 6,
      child: Center(child: Text(title, style: TextStyle(fontSize: 18))),
    ),
  );
}

```

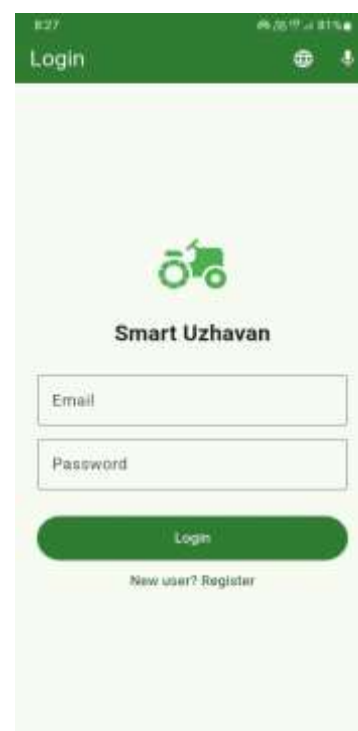

APPENDIX B – SCREENSHOTS



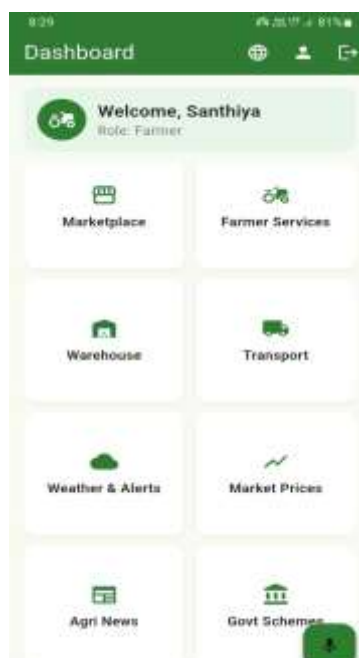
B.1.1 Splash Screen



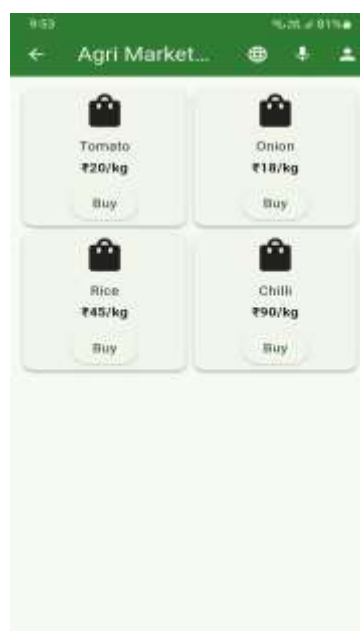
B.1.2 Register Page



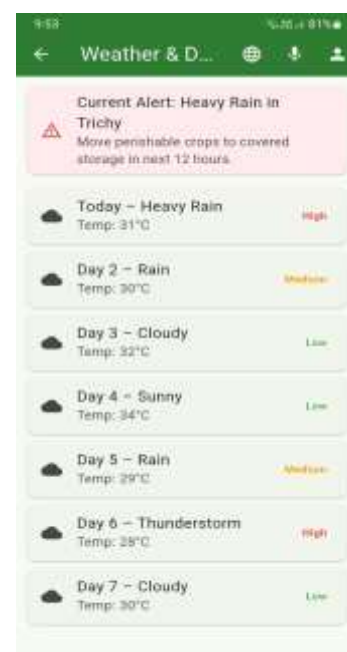
B.1.3 Login Page



B.1.4 Dashboard



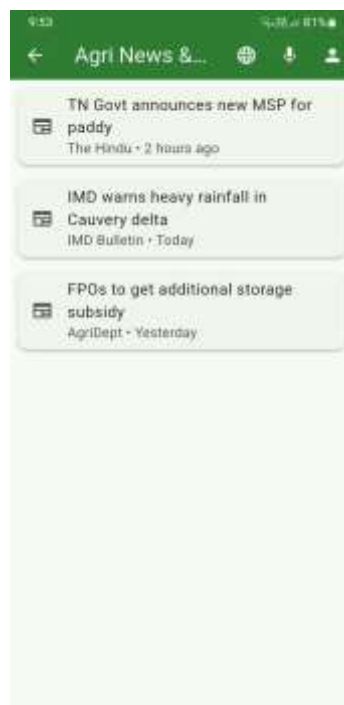
B.1.5 Agri Market



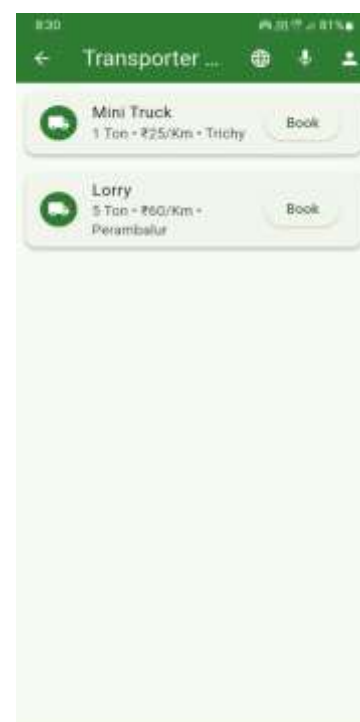
B.1.6 Weather



B.1.7 Farmer Hub



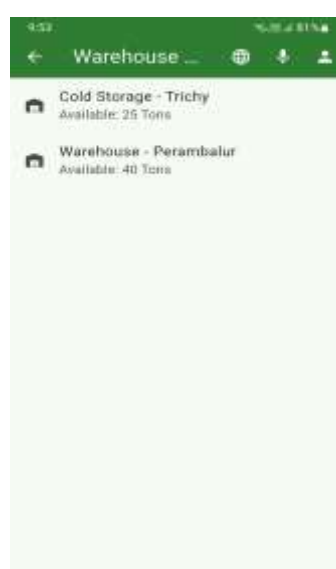
B.1.8 Agri News



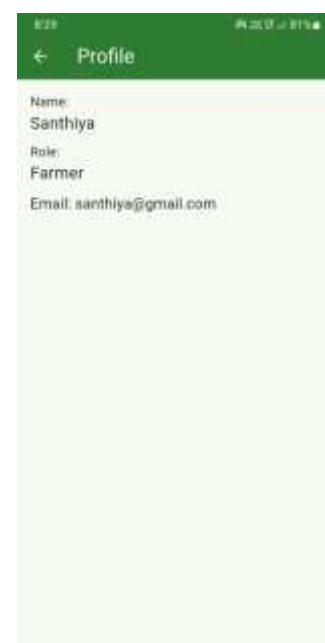
B.1.9 Transporter



B.1.10 Market Price



B.1.11 Warehouse



B.1.12 Profile

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